

Mobile Application Development

Practical No. 22

Develop a registration application to insert and retrieve the data from the database.

Use Database:-

1. activity_main.xml :-

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/etId"
        android:hint="Enter ID (for Update/Delete)"
        android:inputType="number"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etName"
        android:hint="Enter Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etEmail"
        android:hint="Enter Email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etMobile"
        android:hint="Enter Mobile No"
        android:inputType="phone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnInsert"
        android:text="Insert"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnView"
        android:text="View Data"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
```

```

<Button
    android:id="@+id/btnUpdate"
    android:text="Update"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<Button
    android:id="@+id/btnDelete"
    android:text="Delete"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<!-- Display Area -->
<ScrollView
    android:layout_width="match_parent"
    android:layout_height="200dp">

    <TextView
        android:id="@+id/tvData"
        android:text="Data will appear here..."
        android:textSize="16sp"
        android:padding="10dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
</ScrollView>

</LinearLayout>

```

2 - MainActivity.java

```

package com.example.practical22;

import androidx.appcompat.app.AppCompatActivity;
import android.database.Cursor;
import android.os.Bundle;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    EditText etId, etName, etEmail, etMobile;
    Button btnInsert, btnView, btnUpdate, btnDelete;
    TextView tvData;
    DBHelper db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etId = findViewById(R.id.etId);
        etName = findViewById(R.id.etName);
        etEmail = findViewById(R.id.etEmail);
        etMobile = findViewById(R.id.etMobile);

        btnInsert = findViewById(R.id.btnInsert);
        btnView = findViewById(R.id.btnView);
        btnUpdate = findViewById(R.id.btnUpdate);
        btnDelete = findViewById(R.id.btnDelete);
    }
}

```

```

tvData = findViewById(R.id.tvData);

db = new DBHelper(this);

// INSERT
btnInsert.setOnClickListener(v -> {
    boolean result = db.insertData(
        etName.getText().toString(),
        etEmail.getText().toString(),
        etMobile.getText().toString()
    );

    Toast.makeText(this,
        result ? "Inserted Successfully" : "Insert Failed",
        Toast.LENGTH_SHORT).show();
});

// VIEW (DISPLAY BELOW)
btnView.setOnClickListener(v -> {
    Cursor cursor = db.getData();

    if (cursor.getCount() == 0) {
        tvData.setText("No Data Found");
        return;
    }

    StringBuilder data = new StringBuilder();

    while (cursor.moveToNext()) {
        data.append("ID: ").append(cursor.getInt(0)).append("\n");
        data.append("Name:
    ").append(cursor.getString(1)).append("\n");
        data.append("Email:
    ").append(cursor.getString(2)).append("\n");
        data.append("Mobile:
    ").append(cursor.getString(3)).append("\n\n");
    }

    tvData.setText(data.toString());
});

// UPDATE
btnUpdate.setOnClickListener(v -> {
    boolean result = db.updateData(
        etId.getText().toString(),
        etName.getText().toString(),
        etEmail.getText().toString(),
        etMobile.getText().toString()
    );

    Toast.makeText(this,
        result ? "Updated Successfully" : "Update Failed",
        Toast.LENGTH_SHORT).show();
});

// DELETE
btnDelete.setOnClickListener(v -> {
    boolean result = db.deleteData(etId.getText().toString());

    Toast.makeText(this,

```

```

        result ? "Deleted Successfully" : "Delete Failed",
        Toast.LENGTH_SHORT).show();
    });
}
}

```

3 - DBHelper.java

```

package com.example.practical22;

import android.content.Context;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.content.ContentValues;
import android.database.Cursor;

public class DBHelper extends SQLiteOpenHelper {

    public static final String DB_NAME = "UserDB";
    public static final int DB_VERSION = 1;

    public DBHelper(Context context) {
        super(context, DB_NAME, null, DB_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE users (" +
            "id INTEGER PRIMARY KEY AUTOINCREMENT, " +
            "name TEXT, " +
            "email TEXT, " +
            "mobile TEXT)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion)
    {
        db.execSQL("DROP TABLE IF EXISTS users");
        onCreate(db);
    }

    // INSERT
    public boolean insertData(String name, String email, String mobile) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues cv = new ContentValues();

        cv.put("name", name);
        cv.put("email", email);
        cv.put("mobile", mobile);

        long result = db.insert("users", null, cv);
        return result != -1;
    }

    // VIEW
    public Cursor getData() {
        SQLiteDatabase db = this.getReadableDatabase();
        return db.rawQuery("SELECT * FROM users", null);
    }
}

```

```

// UPDATE
public boolean updateData(String id, String name, String email, String
mobile) {
    SQLiteDatabase db = this.getWritableDatabase();
    ContentValues cv = new ContentValues();

    cv.put("name", name);
    cv.put("email", email);
    cv.put("mobile", mobile);

    int result = db.update("users", cv, "id=?", new String[]{id});
    return result > 0;
}

// DELETE
public boolean deleteData(String id) {
    SQLiteDatabase db = this.getWritableDatabase();
    int result = db.delete("users", "id=?", new String[]{id});
    return result > 0;
}
}

```

- **Output:-**

The screenshot shows an Android application interface with a light purple background. At the top, there is a status bar with the time 12:45 and battery level 90%. The main interface consists of four text input fields with placeholder text: "Enter ID (for Update/Delete)", "Enter Name", "Enter Email", and "Enter Mobile No". Below these fields are four purple buttons with white text: "Insert", "View Data", "Update", and "Delete". At the bottom, there is a section displaying user data for two entries. The first entry shows "ID: 5", "Name: Akshay Waghmode", "Email: akshaywaghmode38@gmail.com", and "Mobile: 9527861386". The second entry shows "ID: 6", "Name:", "Email:", and "Mobile:". A home indicator bar is visible at the very bottom.